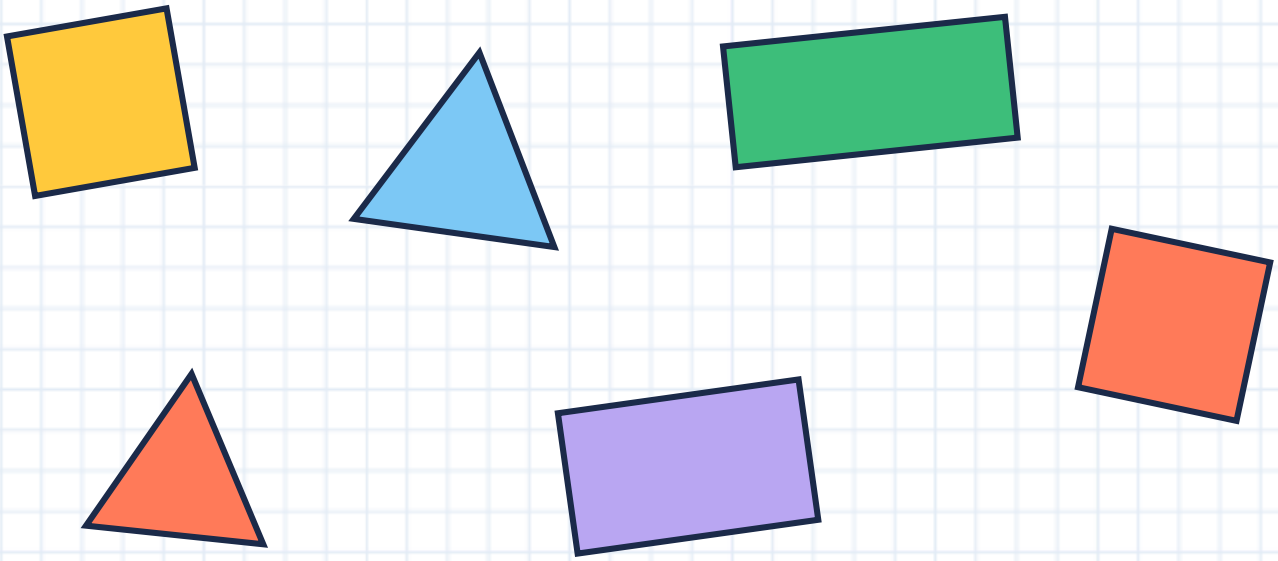


PRIMARY 3 · HOLIDAY PROGRAM

Math Inquirers

Notice it. Test it. Explain why.

Name _____ **Group** _____

My Inquirer Rules

1. Notice the pattern.
2. Test it with objects first.
3. Check your answer.
4. Explain why.

My Inquiry Journal

Your tutor stamps each lesson.

Lesson 1
Number Sleuths

Lesson 2
Cryptarithm Crackers

Lesson 3
Times Table Treasure

Lesson 4
Model Makers

Lesson 5
Remainder Riddles

Lesson 6
Fraction Walls

Lesson 7
Area Architects

Lesson 8
Angle and Line Detectives

Lesson 9
Measure and Money Masters

Lesson 10
Challenge Day

Tutor: _____ Program dates: _____

NUMBER SLEUTHS

Name _____ Group _____ Date _____

Digit cards: use each card once in each number. A 4-digit number cannot start with 0.

No.1

Use these digit cards once each to make 4-digit numbers.

3	0	7	5

a The greatest number: _____

b The smallest number: _____

c The greatest ODD number: _____

d The number closest to 5000: _____

e Why is (d) the closest? _____

No.2

Make a list in order. Write ALL the 4-digit numbers you can make with the cards 2, 0, 5, 5. Start with the smallest.

Starting with 2	
Starting with 5	

a How many numbers did you find? _____

b How many of them are greater than 5200? _____

No.3

Look for a pattern. Write the next two numbers.

a 9850, 9750, 9650, _____, _____

b 1234, 2345, 3456, _____, _____

c 1000, 1001, 1003, 1006, 1010, _____, _____

d 3000, 3250, 3500, 3750, ... The first number greater than 5000 is _____

No.4

Place value puzzles.

a 4 thousands, 15 hundreds, 2 tens and 7 ones make _____

b 7305 = 6 thousands + _____ hundreds + 5 ones

c Mei says that 6 thousands, 0 hundreds and 23 tens is 6023. Is she right? Explain.

.....

.....

No.5

What number am I? Use one clue at a time.

a Read the clues.

> Clue 1: I am between 6000 and 7000.
> Clue 2: My ones digit is half of my thousands digit.
> Clue 3: My hundreds digit is the sum of my other three digits.

I am: _____

b I am a 4-digit ODD number. All my digits are different. My digits add up to 6. What is the greatest number I can be? What is the smallest?

Greatest: _____
Smallest: _____

No.6

How many 4-digit numbers have digits that add up to 3? Make a list in order. Start with the biggest thousands digit.

Thousands digit 3	
Thousands digit 2	
Thousands digit 1	

a How many numbers altogether? _____

b How many start with 3? With 2? With 1? What do you notice?

.....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

Use the cards 5, 0, 8, 2 once each. Make the greatest even number and the smallest odd number.

No.2

How many 4-digit numbers can you make with the cards 1, 1, 0, 0? List them.

No.3

A number has 7 thousands, 13 hundreds and 6 ones. What is the number?

No.4

I am between 5000 and 6000. My hundreds, tens and ones digits are all the same. My digits add up to 11. What number am I?

Challenge (for strong thinkers)

No.5

How many 4-digit numbers have digits that add up to 4? Make a list in order.

No.6

A 4-digit number has all different digits. Its thousands digit is the sum of its other three digits. What is the greatest such number? What is the smallest?

No.7

Mei uses the cards 1, 2, 3, 4 once each to make 4-digit numbers. She lists them all from smallest to greatest. What is the 10th number in her list?

CRYPTARITHM CRACKERS

Name _____ Group _____ Date _____

Work one column at a time, starting with the ones. Remember the carried 1!

No.1

Find the missing digits. Write them in the boxes.

(a)

$$\begin{array}{r} 3\square 47 \\ +\ 25\square 8 \\ \hline \square 025 \end{array}$$

(b)

$$\begin{array}{r} 7\square 03 \\ -\ \square 46\square \\ \hline 2538 \end{array}$$

c Check (b) by adding: 2538 + _____ = _____

No.2

Use the digit cards 1 to 6, each once, to make two 3-digit numbers.

1	2	3	4	5	6

a Make the sum as GREAT as possible. _____ + _____ = _____

b Make the sum as SMALL as possible. _____ + _____ = _____

c Make the DIFFERENCE as small as possible. Guess, check, then improve. Write your tries.

.....

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Best: _____

No.3

Magic square: every row, column and diagonal adds up to the same total.

1200	1700	1600
	1500	

a What is the magic total? _____ Fill in the missing numbers.

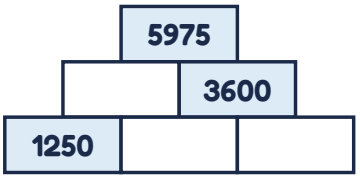
b Look at two numbers opposite each other across the centre. What do you notice?

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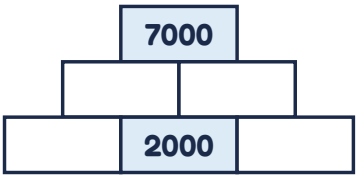
No.4

Number pyramids: each brick is the sum of the two bricks under it. Work backwards.

(a)



(b) corner bricks are equal



No.5

Look for a pattern.

a 4127 + _____ = 9999 and 2593 + _____ = 9999

b What do you notice about the digits? Why is there no carrying?

.....

c Use your pattern: 10 000 – 4127 = _____

No.6

Shape puzzle. Each shape stands for a digit. Different shapes are different digits.



a ★ = _____ ● = _____ ▲ = _____

b How do you know? Start with the ones column.

.....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

Find the missing digits: 2☐7☐ + ☐6☐9 = 6000

No.2

In a number pyramid, each brick is the sum of the two bricks under it. The bottom row is 1200, ☐, ☐. The middle row is 2650, ☐. The top brick is 6000. Find the missing bricks.

No.3

Ali meant to add 1099 to a number, but he added 1999 by mistake and got 6543. What is the correct answer?

No.4

What is the smallest 4-digit number whose digits add up to 30? What is the greatest?

Challenge (for strong thinkers)

No.5

◆◆ + ●● + ▲▲ = ◆●▲. Different shapes are different digits. Find them.

No.6

Use the digit cards 1 to 8 once each to make two 4-digit numbers. Make their difference as small as possible.

No.7

★■● + ★●■ = ●■★. Different shapes are different digits. Find them.

TIMES TABLE TREASURE

Name _____ Group _____ Date _____

No.1

Patterns in the 9 times table. Fill in the answers.

9 ×	1	2	3	4	5	6	7	8	9	10
Answer	9	18								

a Look at the tens digits and the ones digits. Write TWO patterns you notice.

b Ali says 9 × 8 = 81, because 8 + 1 = 9. Use a pattern to explain why he is wrong.

No.2

Last digits in the 7 times table. Write only the LAST digit of each answer.

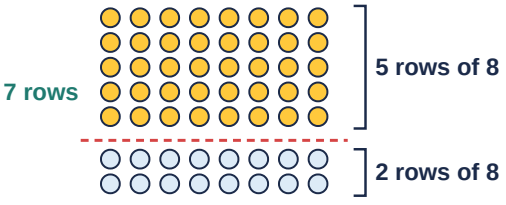
7 ×	1	2	3	4	5	6	7	8	9	10
Last digit	7	4								

a Find ALL the other tables from 2 to 9 that also use all ten last digits. _____

b Without working out 7 × 46, what is its last digit? Explain.

No.3

Mei does not know 7 × 8. She splits 7 rows of 8 dots into two parts.



a 5 × 8 = _____ 2 × 8 = _____ so 7 × 8 = _____

b Split 9 × 6 into two facts you know. Show your working.

No.4

Grid puzzle. Each number in the grid is the row number × the column number. The row and column numbers are all from 1 to 10. Find the six missing row and column numbers, then fill in the grid.

×			
	56		
			81
	48	**36**	

Hint: which number can be made in only one way?

No.5

Ben puts 36 chairs in equal rows. There must be at least 2 rows and at least 2 chairs in each row. 3 rows of 12 is different from 12 rows of 3.

a Find ALL the ways. Try 2 rows, then 3 rows, then 4 rows … in order.

Number of rows	2							
Chairs in each row	18							

b How many ways are there? _____

c Most ways come in pairs, like 2 rows of 18 and 18 rows of 2. Which way has no partner? Why?

.....

.....

No.6

Number riddle.

a Find ALL the 2-digit numbers that can be divided exactly by 6 AND by 8. Hint: list the 2-digit numbers you can divide exactly by 8 (16, 24, 32 …), then test each one.

.....

.....

b I am one of your numbers. My digits add up to 12. What number am I?

Answer: _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

The digits of every answer in the 9 times table (9 × 1 to 9 × 10) add up to 9. Which of these are in the 9 times table: 54, 65, 72, 83?

No.2

What is the last digit of 8 × 8 × 8? Use last digits only.

No.3

Ravi says: "Every answer in the 6 times table is even." Is he right? Explain why.

No.4

Find ALL the 2-digit numbers that can be divided exactly by 7 and whose digits add up to 10.

Challenge (for strong thinkers)

No.5

Mei has between 40 and 60 sweets. She can share them equally among 6 friends with none left over. She can also share them equally among 9 friends with none left over. How many sweets does she have?

No.6

Ben keeps multiplying by 7: 7, 49, 343 ... He writes only the last digit each time: 7, 9, 3 ... What is the last digit when he has multiplied 8 sevens together? Explain the pattern.

No.7

I am a 3-digit number less than 200. I can be divided exactly by 7 and by 9. My digits add up to 9. What number am I?

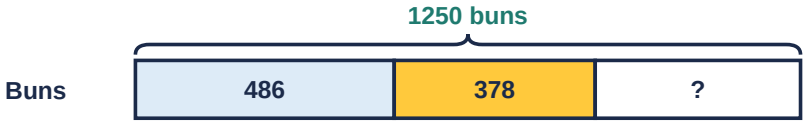
MODEL MAKERS

Name _____ Group _____ Date _____

Draw a model. Label what you know. Put a ? on what you want to find.

No.1

A bakery made 1250 buns. It sold 486 buns in the morning and 378 buns in the afternoon.

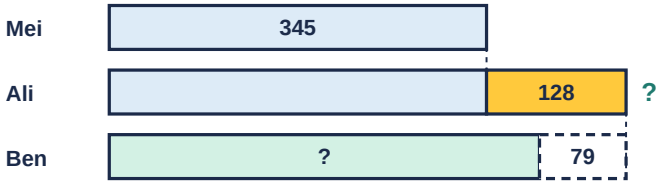


a How many buns were left? _____

b How many more buns were sold in the morning than in the afternoon? _____

No.2

Mei has 345 stickers. Ali has 128 more stickers than Mei. Ben has 79 fewer stickers than Ali.



a Ali has _____ stickers. Ben has _____ stickers.

b How many stickers do the three children have altogether? _____

No.3

Siti has 3 times as many beads as Ravi. They have 248 beads altogether.



a How many units are there altogether? _____ 1 unit = _____ beads

b Siti has _____ beads. She has _____ more beads than Ravi.

No.4

Ali and Ben have 150 marbles altogether. Ali has 24 more marbles than Ben. How many marbles does each boy have?

Draw your model here.

Ben: _____ Ali: _____

No.5

Mei has 85 stickers. Siti has 47 stickers. Mei gives some stickers to Siti so that they both have the same number.

Draw your model here.

a How many stickers does Mei give Siti? _____

b Ravi says, "Mei must give Siti 38 stickers, because 85 – 47 = 38." What mistake did Ravi make?

.....

.....

No.6

Ali has $45. Ben has $18. Grandma gives each boy the same amount of money. Now Ali has twice as much money as Ben.

Draw your model here.

a What stays the same before and after Grandma gives the money? _____

b How much money did Grandma give each boy? _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

Ali and Ben have 96 cards altogether. Ali has 3 times as many cards as Ben. How many more cards does Ali have than Ben?

No.2

Siti has 54 more beads than Mei. Siti gives Mei some beads, and now they have the same number. How many beads did Siti give Mei?

No.3

Three boxes hold 600 pens altogether. Box A has 50 more pens than Box B. Box C has 50 more pens than Box A. How many pens are in Box C?

No.4

Mei is 8 years old. Her mother is 36. How old will Mei be when her mother is 3 times as old as Mei?

Challenge (for strong thinkers)

No.5

Ravi is 7 years old and his sister is 3. In how many years will their ages add up to 30?

No.6

Ali has $60 more than Ben. Ali spends $15 and Ben spends $35. Now Ali has 5 times as much money as Ben. How much money did Ben have at first?

No.7

A ribbon 4 m 75 cm long is cut into 3 pieces. The second piece is twice as long as the first. The third piece is 25 cm longer than the second. How long is the third piece?

REMAINDER RIDDLES

Name _____ Group _____ Date _____

When we share or make groups, some may be left over. The leftover is called the remainder. It is always smaller than the number we divide by.

No.1

Siti shares 47 stickers equally among 6 friends.

a How many stickers does each friend get? _____ How many are left over? _____

b What is the smallest number of extra stickers Siti needs so that none are left over? _____

c Ben works out 60 ÷ 7. He writes: 60 ÷ 7 = 7 remainder 11. What is wrong? Write the correct answer.

No.2

Divide. Then decide what to do with the leftover.

a 75 pupils go on a trip. A minibus holds 9 pupils. How many minibuses are needed? _____

b Mei has $60. A book costs $7. How many books can she buy? _____

c A farmer packs 87 eggs into boxes of 6. How many boxes can he fill completely? _____ How many boxes does he need to hold ALL the eggs? _____

No.3

Ravi packs marbles into bags of 8. He fills 12 bags and has 5 marbles left over.

a How many marbles does Ravi have? _____

b He tips them all out and packs them again into bags of 9. How many full bags does he get? How many marbles are left over?

Full bags: _____ Left over: _____

No.4

What number am I? Make a list, then test each number.

a I am between 40 and 60. When I am divided by 5, the remainder is 3. When I am divided by 4, the remainder is 1.

Remainder 3 when ÷ 5				
Remainder when ÷ 4				

I am: _____

b I am a 2-digit number less than 50. When I am divided by 6, the remainder is 5. When I am divided by 4, the remainder is 3. Find ALL the numbers I could be.

.....

.....

No.5

Dividing by 8.

a Write ALL the remainders you could get when you divide a number by 8.

Answer: _____

b A number between 60 and 70 is divided by 8. The remainder is as big as it can be. What is the number? _____

No.6

At a class party, children sit at tables. If 6 children sit at each table, 5 children have no seat. If 7 children sit at each table, every seat is filled and nobody is left over.

Number of tables	1	2	3	4	5	6
Children: 6 at each table, 5 with no seat	11					
Children: 7 at each table	7					

a How many tables are there? _____ How many children? _____

b If 8 children sit at each table instead, how many seats are empty? _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

Mei has 53 beads. She makes bracelets with 8 beads each. How many bracelets can she make? How many more beads does she need to make one more bracelet?

No.2

I am a number between 20 and 30. When I am divided by 4, the remainder is 3. When I am divided by 6, the remainder is 5. What number am I?

No.3

What is the biggest 2-digit number that leaves a remainder of 8 when it is divided by 9?

No.4

A boat can carry 9 people. 100 people want to cross the river. How many trips must the boat make?

Challenge (for strong thinkers)

No.5

A number is divided by 6. The answer and the remainder are the same number, and it is not 0. Find all the numbers it could be.

No.6

Today is Monday. What day of the week will it be 100 days from today?

No.7

Mr Lim gives pens to his pupils. If each pupil gets 4 pens, 10 pens are left over. If each pupil gets 6 pens, he is 4 pens short. How many pupils are there? How many pens does he have?

FRACTION WALLS

Name _____ Group _____ Date _____

Each row of a fraction wall is one whole, cut into EQUAL parts. To compare or add fractions, make the parts the same size.

No.1

Use the fraction wall.

1 whole											
$\frac{1}{2}$						$\frac{1}{2}$					
$\frac{1}{3}$				$\frac{1}{3}$				$\frac{1}{3}$			
$\frac{1}{4}$			$\frac{1}{4}$			$\frac{1}{4}$			$\frac{1}{4}$		
$\frac{1}{6}$		$\frac{1}{6}$			$\frac{1}{6}$			$\frac{1}{6}$			$\frac{1}{6}$
$\frac{1}{8}$		$\frac{1}{8}$		$\frac{1}{8}$		$\frac{1}{8}$		$\frac{1}{8}$		$\frac{1}{8}$	
$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$

a $\frac{1}{2} = \frac{\square}{4} = \frac{\square}{6} = \frac{\square}{8} = \frac{\square}{12}$

b $\frac{2}{3} = \frac{\square}{6} = \frac{\square}{12}$ (c) $\frac{3}{4} = \frac{\square}{8} = \frac{\square}{12}$

d Look at your answers to (a). What do you notice about the top and bottom numbers?

.....

No.2

Write >, < or = in each circle.

(a) $\frac{1}{5} \bigcirc \frac{1}{7}$	(b) $\frac{3}{8} \bigcirc \frac{5}{8}$	(c) $\frac{2}{3} \bigcirc \frac{4}{6}$
(d) $\frac{3}{4} \bigcirc \frac{5}{8}$	(e) $\frac{4}{9} \bigcirc \frac{4}{7}$	(f) $\frac{2}{3} \bigcirc \frac{3}{4}$

g Put in order, smallest first: $\frac{5}{12}$, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{3}{4}$

Smallest first: _____

No.3

Each shape is cut into equal parts.



A B C D E

a Which shapes have exactly $\frac{1}{4}$ shaded? _____

b Write the fraction shaded in each of the other shapes. _____

No.4

Add or subtract. Make the parts the same size first.

(a) $\frac{2}{7} + \frac{3}{7} =$ _____	(b) $\frac{1}{4} + \frac{3}{8} =$ _____	(c) $\frac{1}{3} + \frac{5}{12} =$ _____
(d) $\frac{5}{6} - \frac{1}{3} =$ _____	(e) $1 - \frac{3}{10} =$ _____	(f) $\frac{1}{2} + \frac{3}{10} =$ _____

No.5

Ravi paints $\frac{1}{6}$ of a fence in the morning and $\frac{2}{3}$ of it in the afternoon. Draw a bar.

a What fraction has he painted? _____ (b) What fraction is not painted? _____

c How much more did he paint in the afternoon than in the morning? _____

No.6

Partners that make ONE WHOLE.

Pair each yellow card with the blue card that makes one whole with it. Write the 5 pairs.

$\frac{1}{4}$	$\frac{7}{12}$	$\frac{2}{3}$	$\frac{6}{8}$	$\frac{3}{8}$	$\frac{5}{6}$	$\frac{5}{12}$	$\frac{5}{8}$	$\frac{1}{6}$	$\frac{4}{12}$

.....

No.7

Find ALL the fractions with bottom number 12 that are bigger than $\frac{1}{2}$ but smaller than $\frac{3}{4}$.

a _____ (b) Which one is equal to $\frac{2}{3}$? _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

Which is bigger, 3/5 or 7/10? Show how you know.

No.2

Siti reads 1/4 of a book on Monday and 1/2 of the book on Tuesday. What fraction of the book has she read? What fraction is left?

No.3

Put these in order, smallest first: 2/3, 5/6, 7/12.

No.4

Write two fractions equal to 4/6: one with a smaller bottom number and one with a bigger bottom number.

Challenge (for strong thinkers)

No.5

A fraction is equal to 2/3. Its bottom number is 4 more than its top number. What is the fraction?

No.6

Fill one whole bar using exactly 5 pieces. You may use 1/2, 1/4 and 1/8 pieces, as many of each as you like. Find all the ways.

No.7

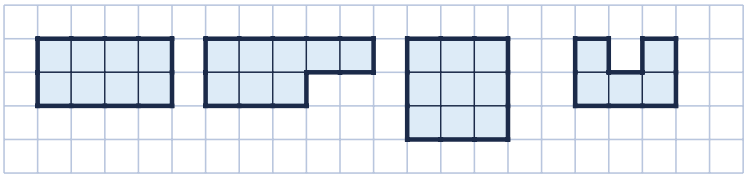
Siti and Ravi each have the same size bottle of juice. Siti drinks 2/3 of her bottle. Ravi drinks 7/12 of his. Who drinks more? How much more?

AREA ARCHITECTS

Name _____ Group _____ Date _____

No.1

Each small square is 1 cm by 1 cm. Find the area (in cm²) and the perimeter (in cm) of each figure.



A B C D

Figure	A	B	C	D
Area (cm²)				
Perimeter (cm)				

a Which two figures have the same area but different perimeters? _____

b Which figures have the same perimeter but different areas? _____

No.2

Siti has 18 m of fence. She makes a rectangular garden with the whole fence. Each side is a whole number of metres. List ALL the gardens she can make, in order.

Length (m)	Breadth (m)	Check: perimeter	Area (m²)

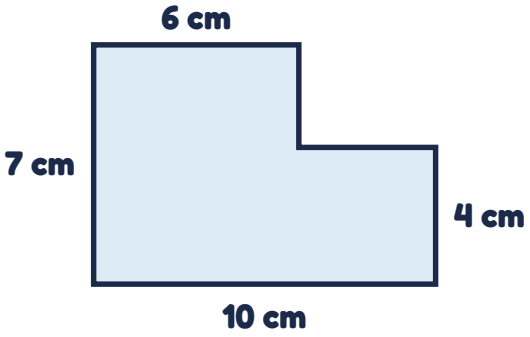
a Which garden has the biggest area? _____

No.3

A square has the same perimeter as a rectangle 7 cm long and 3 cm wide. What is the area of the square? _____ cm²

No.4

Find the two missing sides first. Then find the area and the perimeter of this figure.



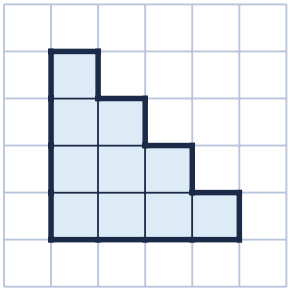
Missing sides: _____ and _____

a Area = _____ cm^2

b Perimeter = _____ cm

No.5

Look at this staircase.



a What is its area? _____ cm^2

b What is its perimeter? _____ cm

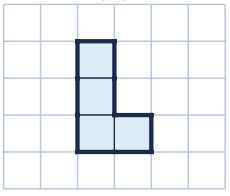
c Ben fills in the gaps with squares to make a 4 by 4 square. What happens to the perimeter? What happens to the area?

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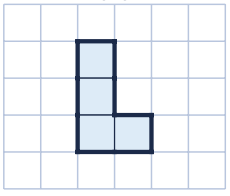
No.6

This L shape is made of 4 squares. Its perimeter is 10 cm. Add ONE square that shares a whole side with the L. Shade it.

(a)



(b)



a In grid (a), add a square so that the perimeter becomes 12 cm.

b In grid (b), add a square so that the perimeter stays 10 cm.

c Can you add one square so that the perimeter becomes 8 cm? Explain.

.....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

Find all the rectangles with whole-cm sides and an area of 24 cm². Which one has the shortest perimeter?

No.2

A rectangle is 12 cm long. Its perimeter is 30 cm. Find its area.

No.3

A rectangle 8 cm by 2 cm and a square have the same perimeter. Which has the bigger area, and by how much?

No.4

A square with sides of 6 cm is cut into two rectangles of the same size. What is the perimeter of each rectangle?

Challenge (for strong thinkers)

No.5

A rectangle is made of 3 same-size squares in a row. Its perimeter is 32 cm. What is its area?

No.6

Ali joins 5 same-size squares with sides of 2 cm, whole sides touching. What is the biggest perimeter his shape can have? The smallest?

No.7

A rectangle 7 cm by 5 cm has a 1 cm square cut out of each of its 4 corners. Find the new area and the new perimeter.

ANGLE AND LINE DETECTIVES

Name _____ Group _____ Date _____

No.1

Fold a right-angle checker. Use it to sort the angles a to f.

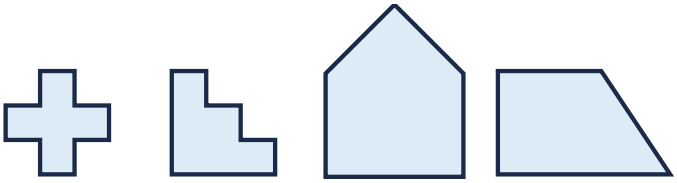


a b c d e f

Smaller: _____ Right angle: _____ Bigger: _____

No.2

How many right angles are INSIDE each shape? Check every corner with your checker.

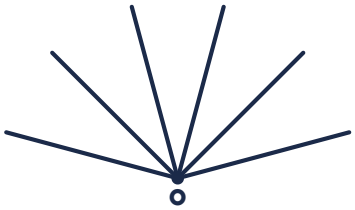


A B C D

Shape	A	B	C	D
Right angles inside				

No.3

These 6 lines all start at point O.



O

a How many angles? Count the small ones first, then those made of 2 small ones, and so on.

Made of 1: ____ 2: ____ 3: ____ 4: ____ 5: ____

Total: _____

b Look for a pattern. 8 lines start at one point, spread out like these. How many angles are there? _____

No.4

Look at the lines in these letters. Write Yes or No in the table.

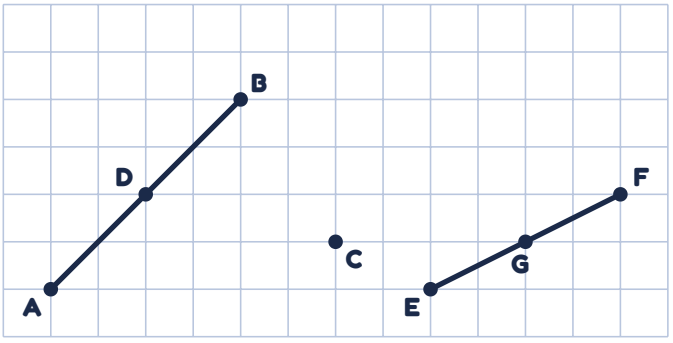
E H T L Z N

Letter	E	H	T	L	Z	N
Parallel lines?						
Perpendicular lines?						

b In the letter E: pairs of parallel lines _____ pairs of perpendicular lines _____

No.5

Use a ruler. Draw on the grid.



a Through C, draw a line parallel to AB. (b) Through D, draw a line perpendicular to AB.

c Through G, draw a line perpendicular to EF. Check it with your checker.

No.6

How many right angles can you find in this window?



Visit every point where lines meet. Write how many right angles are at each point, then add.

Right angles altogether: _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

I have 4 sides and 4 right angles. My sides are not all the same length. What shape am I? How many pairs of parallel lines do I have?

No.2

A U shape is made of 5 squares: 3 squares in a row with 1 more square on top of each end square. How many right angles are inside it?

No.3

At which o'clock times do the hands of a clock make a right angle?

No.4

The # sign is made of 4 lines. How many pairs of parallel lines does it have? How many pairs of perpendicular lines?

Challenge (for strong thinkers)

No.5

A 4-sided shape has 3 right angles. What can you say about its 4th angle? Can a 5-sided shape have exactly 3 right angles? Draw one.

No.6

A long rectangle is cut into 6 squares in a row by 5 lines. How many right angles can you find?

No.7

Can a 6-sided shape have 6 right angles inside it? What is the most right angles a 6-sided shape can have inside?

MEASURE AND MONEY MASTERS

Name _____ Group _____ Date _____

No.1

Trick 1: change to ONE unit. 1 m = 100 cm 1 kg = 1000 g 1 ℓ = 1000 ml $1 = 100¢

a 3 m 6 cm = _____ cm (b) 2 kg 450 g = _____ g

c 4080 ml = _____ ℓ _____ ml (d) 1 m 75 cm + 85 cm = _____ m _____ cm

No.2

Four bags of rice are weighed: 1 kg 300 g 1030 g 1 kg 1 kg 3 g

a Order them, lightest first: _____, _____, _____, _____

b Ben says, "1 kg 30 g is the same as 130 g." Explain his mistake.

.....

No.3

Mei has a ribbon 3 m long. She cuts 4 pieces from it. Each piece is 65 cm long.

a How much ribbon does she cut off altogether? _____ m _____ cm

b How much ribbon is left? _____ cm

c Can she cut one more 65 cm piece? If not, how much more ribbon does she need?

_____

No.4

Siti has $20. She wants to buy 3 different things from this list.

Book	Pen set	Puzzle	Stickers	Water bottle
$7.85	$4.60	$6.90	$2.35	$5.50

a Can she buy the book, the puzzle and the water bottle? Show how you know.

_____

b Which 3 things should she buy to spend as much of her $20 as she can? How much change will she get?

.....

No.5

The 24-hour clock. After 12 noon, add 12 to the hour.

7:15 a.m.	3:40 p.m.	_____	12:30 p.m.
_____	_____	**20:05**	_____

a Fill in the table. The top row is 12-hour time; the bottom row is 24-hour time.

b Siti says, "17:00 is 7 p.m." Is she right? Explain.

No.6

Trick 2: draw a time line and jump to the next o'clock.

a Ali's swimming lesson starts at 3:45 p.m. and ends at 5:10 p.m. How long is it? _____ h _____ min

b Mei's piano lesson starts at 16:35 and lasts 50 minutes. When does it end? Use 24-hour time. _____

c A film lasts 1 h 50 min. It ends at 21:15. When did it start? _____

No.7

Buses leave the interchange every 25 minutes. The first bus leaves at 06:40.

1st bus	2nd bus	3rd bus	4th bus	5th bus	6th bus
06:40					

a Ravi reaches the interchange at 08:15. When is the next bus? How long does he wait?

Answer: _____

b The bus ride to school takes 35 minutes. When does Ravi reach school? _____

c Ravi wants to reach school by 08:30. Which is the latest bus he can take? _____

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

A 2 ℓ bottle of juice is poured into cups. Each cup holds 300 ml. How many cups can be filled? How much juice is left over?

No.2

A shop opens at 09:30 and closes at 21:00. How long is it open?

No.3

A box with 4 books weighs 1 kg 50 g. The empty box weighs 250 g. All the books weigh the same. How heavy is one book?

No.4

Ben has only 20¢ and 50¢ coins. He has 9 coins worth $3 altogether. How many of each coin does he have?

Challenge (for strong thinkers)

No.5

A bottle full of water weighs 1 kg 200 g. After half of the water is poured out, it weighs 750 g. How heavy is the empty bottle?

No.6

Mei buys 3 pens and 2 files for $9.20. Each file costs 60¢ more than each pen. How much does one pen cost?

No.7

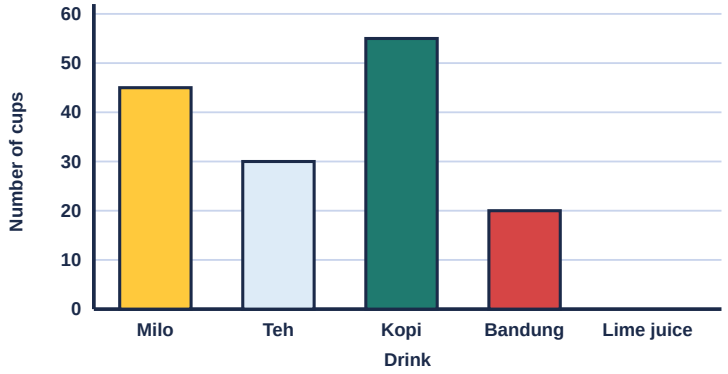
A clock is 10 minutes fast. A train leaves when this clock shows 13:05. The trip takes 1 h 40 min. What is the real time when the train arrives? Give it in 24-hour and 12-hour time.

CHALLENGE DAY

Name _____ Group _____ Date _____

No.1

A drinks stall sold 200 cups of drinks on Monday. The bar graph shows the cups sold, but the bar for Lime juice is missing.



a Which drink shown sold the most? _____ (b) How many more cups of Kopi than Teh? _____

c How many cups of Lime juice were sold? _____ Now draw its bar on the graph.

d Which two drinks together sold exactly 100 cups? _____ and _____

No.2

Ali, Mei, Ravi and Siti are each in a different CCA. Use the clues: put a ✓ or a cross in each box, then write each child's CCA in the last column.

Clue 1 Ali is not in the choir and does not play chess.

Clue 2 Mei and Ravi went to watch their friend sing in the choir concert.

Clue 3 The swimmer's name has 4 letters.

	Chess	Choir	Football	Swimming	CCA
Ali					
Mei					
Ravi					
Siti					

No.3

Draw it! Queues.

a Ben is 3rd from the front and 5th from the back. How many in the queue? _____

b Ali is 2nd from the front, Siti is last, 3 people are between them. How many in all? _____

No.4

List it! Make 3-digit numbers with these cards. Use each card once in every number.

2	5	8

a List ALL the numbers, in order from smallest to largest.

.....

b How many of them are odd numbers? _____

c Which number is closest to 600? _____

No.5

Make a table! Squares are made in a row with sticks. Fill in the table.



square: 4 sticks 2 squares: 7 sticks 3 squares: 10 sticks

Squares	1	2	3	4	5	6
Sticks	4	7	10			

a How many sticks are needed for 8 squares? _____

b Ben uses 31 sticks. How many squares does he make? _____

No.6

I am a 3-digit odd number. My digits add up to 6. My hundreds digit is bigger than my tens digit. None of my digits is 0. What could I be? Find ALL the answers.

.....

.....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Stretch

No.1

In a queue, there are 4 people in front of Ravi and 7 people behind him. How many people are in the queue?

No.2

Ali, Ben and Siti each have a different pet: a cat, a dog and a fish. Ali's pet has no fur. Ben does not have the dog. Who has which pet?

No.3

Triangles are made in a row with sticks: 1 triangle uses 3 sticks, 2 triangles use 5 sticks, 3 triangles use 7 sticks. How many sticks are needed for 12 triangles?

No.4

Use the digits 3, 0, 6 and 9 once each. Make the largest even 4-digit number and the smallest odd 4-digit number.

Challenge (for strong thinkers)

No.5

One of Ali, Ben and Mei ate the last kueh. Ali says, “Ben ate it.” Ben says, “I did not eat it.” Mei says, “I did not eat it.” Only ONE of them is telling the truth. Who ate the kueh?

No.6

In a queue, Siti is 6th from the front and Ravi is 4th from the back. There are 3 children between them, and Siti is in front of Ravi. How many children are in the queue?

No.7

A bar graph shows the paper collected by 4 classes. 3P1 collected 10 kg more than 3P2. 3P3 collected the same as 3P1. 3P4 collected 5 kg less than 3P2. Altogether they collected 135 kg. How much did each class collect?

Cut-outs

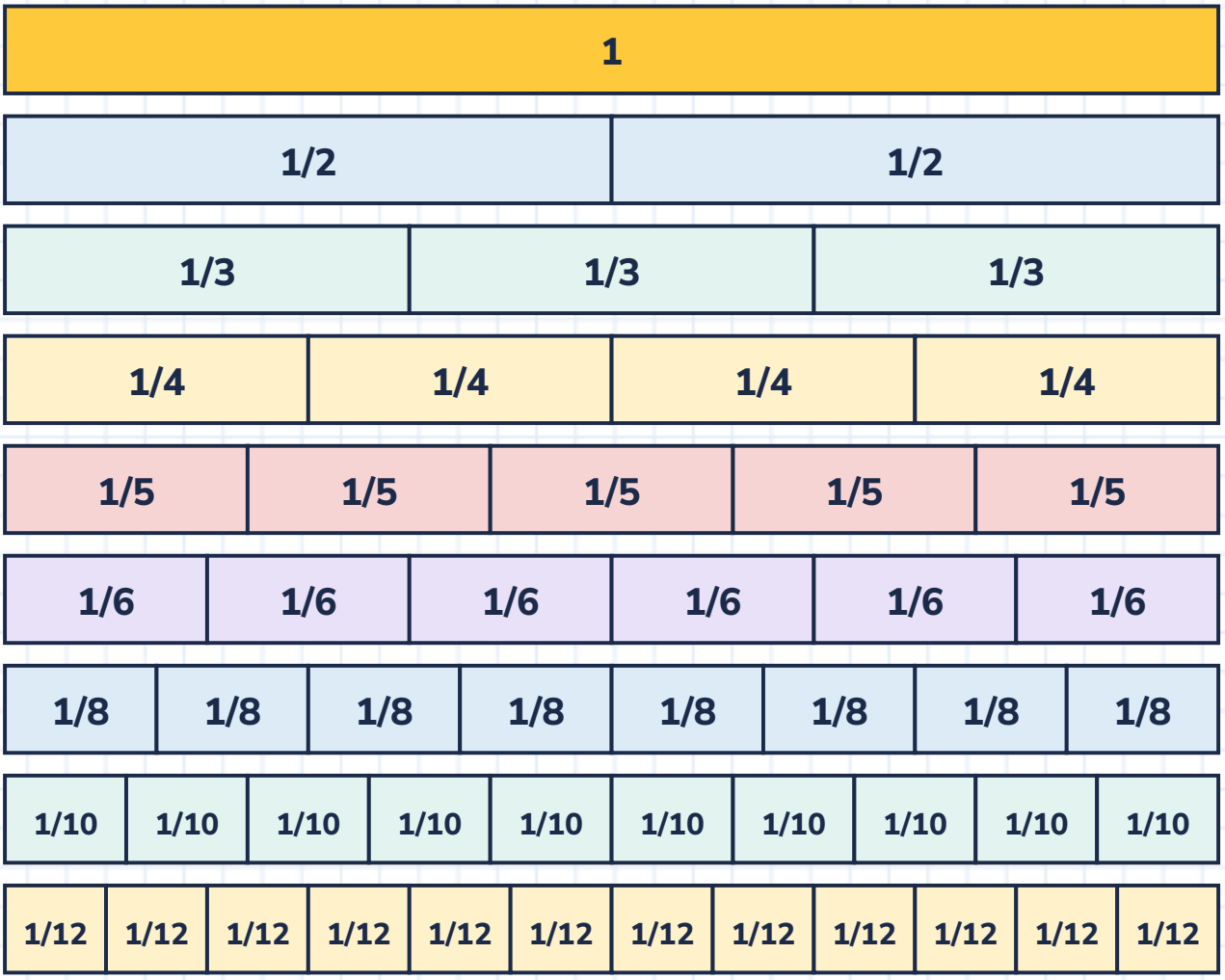
For the P3 program. Cut along the dashed lines and keep the pieces in one envelope.

Digit cards (Lessons 1 and 2)

0	1	2	3	4
5	6	7	8	9
0	1	2	3	4
5	6	7	8	9

Fraction wall

For Lesson 6. Cut the strips apart.

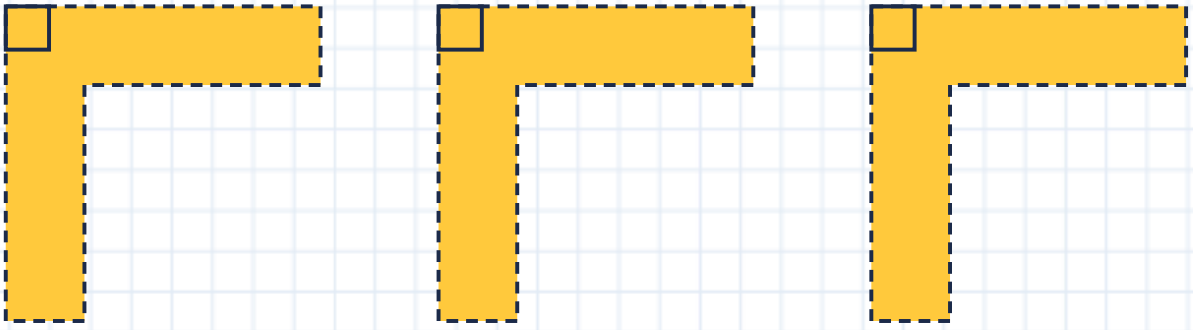


Line the strips up under the whole. Which fractions are the same size?

Square grid

For Lesson 7. Each square is one unit.

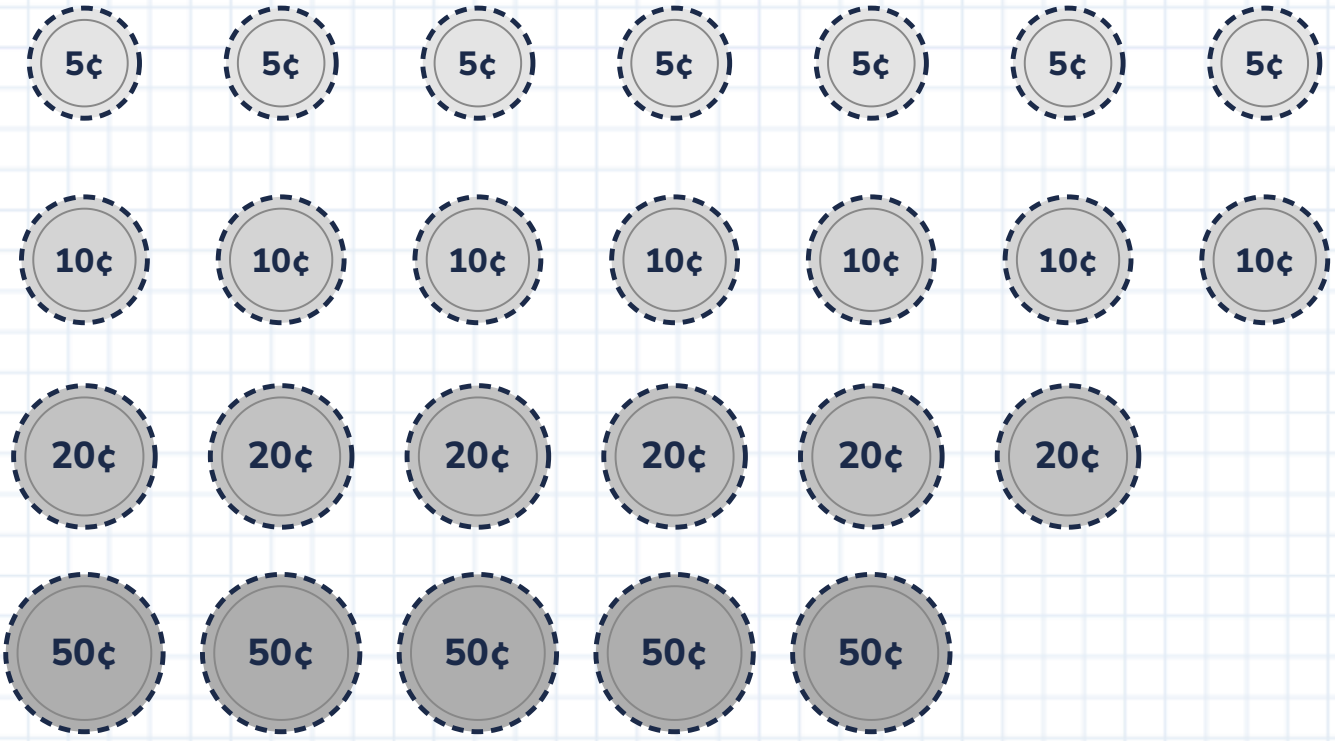
Right-angle testers (Lesson 8)



Cut out a right-angle tester. Put its corner on an angle: bigger, smaller or exactly a right angle?

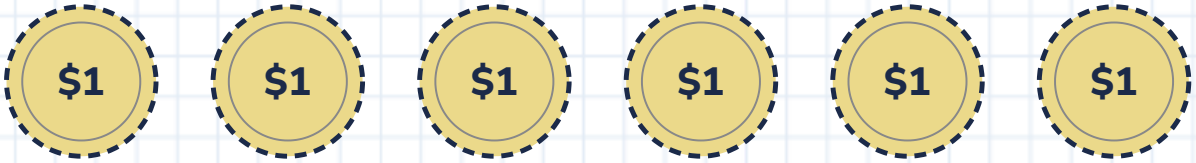
Play coins

For Lesson 9.



Play notes and $1 coins

For Lesson 9.





CERTIFICATE OF COMPLETION

Math Inquirers Holiday Program

Primary 3 Holiday Program

This certifies that

.....

completed all 10 lessons, solved puzzles, spotted patterns and explained their thinking.

Notice it. Test it. Explain why.

Tutor: _____ Date: _____